



Inhalation of *Acori Tatarinowii* Rhizoma essential oil alleviates dyskinesia in Parkinson's disease rats through the regulation of neuroinflammation

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ABSTRACT

Ethnopharmacological relevance: *Acorus tatarinowii* Rhizoma, a traditional Chinese medicine known for opening the orifices and transforming phlegm, is used in the treatment of brain disorders. It is listed as the top grade in the famous herbal monograph Shennong Materia Medica Classic. Traditional Chinese medicine believes that *Acorus tatarinowii* Rhizoma has a good advantage in the treatment of nervous system diseases, and modern research has also found that the essential oil of *Acorus tatarinowii* Rhizoma is the main component that plays a neuroprotective role and plays an important role in the treatment of Parkinson's disease.

Aim of the study: This study aims to explore the effects and mechanisms of essential oil of *Acorus tatarinowii* Rhizoma (EOAT) on LPS-induced BV2 cell damage and Rotenone-induced Parkinson's disease (PD) rat models. **Materials and methods:** In this experiment, the components of EOAT were identified by GC-MS. LPS was used to induce the overactivation of BV2 microglia, and rotenone was injected subcutaneously to induce Parkinson's disease in rats. Then, the expression of inflammatory factors and IBA-1 in cell was evaluated, and the effects of EOAT treatment were assessed on motor function, inflammatory factors, neurotransmitters, TH, α -Syn, and pathways and inflammation-related mRNA in rats.

Results: GC-MS analysis obtained 24 components, among which β -Asarone and α -Asarone had the highest contents. In vitro experiments showed that after 2 h of EOAT intervention, the inflammatory factors TNF- α and IL-6 in the supernatant of LPS-induced BV2 cells were significantly reduced. The IF results showed that after EOAT intervention, the expression of IBA-1 protein in BV2 cells was significantly reduced. In animal experiments, rotenone injection in model rats led to a decrease in motor function, while inhalation of EOAT improved the motor ability of Parkinson's rats. In addition, Madopar and EOAT inhalation increased the levels of BDNF and DA in brain tissue and reduced the levels of IL-1 β , TNF- α , and IL-6. IHC, IF, and WB analyses showed that the expression of TH protein in brain tissue of the Madopar group and the EOAT group was significantly increased, and the expression level of α -syn was reduced. RT-qPCR results showed that compared with the Model group, the levels of *TLR2*, *MyD88*, *NF- κ B*, *IL-1 β* , *TNF- α* , *α -syn*, and *Bax* in the substantia nigra and striatum of the Madopar group and the EOAT group were significantly down-regulated, and the levels of *TH* and *Bcl-2* were significantly up-regulated.

Conclusion: These findings suggest that EOAT can prevent LPS from damaging BV2 cells and significantly improve the motor function of PD rats and lessen neuroinflammation. The anti-Parkinson's mechanism of EOAT is primarily dependent on the regulation of the TLR2/MyD88/NF- κ B signaling pathway.

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1. Introduction

Parkinson's disease (PD) is a prevalent chronic and progressive neurodegenerative disorder (Chen et al., 2022). Its pathology is characterized by the degeneration of dopaminergic neurons in the substantia nigra and the formation of Lewy bodies. It is accompanied by motor symptoms such as resting tremor, bradykinesia, rigidity, and postural instability (Tansey et al., 2022; Rocha et al., 2018). Physical activity, surgery, and pharmacological treatments can enhance the quality of life for those with PD (Nayab et al., 2023). However, these existing treatments only provide symptom relief and do not effectively slow down or completely cure the disease.

Modern medicine attributes PD to basal ganglia malfunction, particularly the degeneration of dopaminergic neurons in the substantia nigra (SN) of the midbrain, which disrupts signal transmission to the striatum, leading to dysfunction in both areas (Ye et al., 2023). PD is a complex disease caused by multiple factors such as genetic factors, personal and environmental factors, aging and so on. Its pathogenesis covers many aspects, including neuroinflammation, α -synuclein (α -syn) accumulation, mitochondrial dysfunction, oxidative stress and excessive accumulation of neuromelanin. These factors act independently of each other and promote each other, leading to the dysfunction and loss of dopaminergic neurons and to characteristic somatic motor disorders and nonmotor symptoms (Liu et al., 2022; Paillusson et al., 2017). Among them, neuroinflammation may be an important mediator in the pathogenesis and progression of the disease (Morris et al., 2024). Pathological chronic neuroinflammation associated with PD is mostly induced by microglia (Ibarra-Gutiérrez et al., 2023), which can trigger elevated levels of various inflammatory molecules and proteins, ultimately leading to cell death (Tansey et al., 2022). Inflammatory substances have the ability to breach the blood-brain barrier and access the brain, leading to neuronal damage and apoptosis by amplifying immune responses and subsequently triggers microglia activation (Liu et al., 2022; Ibarra-Gutiérrez et al., 2023). In addition, neuroinflammation also affects the synthesis and release of neurotransmitters, which leads to the core symptoms of PD, such as movement disorders.

Numerous studies have shown that abnormal accumulation of α -syn is one of the core pathogenic mechanisms of the disease (Tsai et al., 2020). Research indicates that heightened levels of α -syn can lead to harm to mitochondrial structure and function, as well as increase the generation of pro-inflammatory factors such as IL-1 β , TNF- α and others. Pathological α -syn impacts both neurons and glial cells like microglia and astrocytes. Research has shown that α -syn oligomers mostly bind to toll-like receptors (TLRs) to initiate inflammatory reactions. TLRs, also known as immune sentinels, belong to the type I transmembrane receptor family. As pattern recognition receptors, they have the potential to identify pathogens and tissue damage, and can play a part in the activation process of inflammatory response (Li et al., 2018; Farina et al., 2005; Jack et al., 2005; Carty and Bowie, 2011). TLRs activate innate immune cells by recognizing a variety of different pathogen components, thereby triggering an inflammatory response and further promoting the activation of adaptive immunity to effectively eliminate pathogens in the body (Heinz et al., 2020).

Multiple studies have demonstrated that α -syn oligomers secreted by neurons with specific conformations act as agonists of TLR2 and can trigger inflammatory responses in microglia (Kim et al., 2013). TLR2 is a pathological α -syn receptor that mediates neuroinflammation in PD and leads to neurodegenerative changes that ultimately lead to disease progression. In addition, Dutta et al. found that the TLR2/MyD88/NF- κ B pathway is crucial in the spread of α -syn. Inhibiting this mechanism by deleting TLR2 gene or using peptides can decrease α -syn spread and safeguard dopaminergic neurons (Dutta et al., 2021). Cheng et al. verified that the TLR2/MyD88/NF- κ B pathway was suppressed in the prodromal PD rat model, resulting in reduced levels of inflammatory factors and enhanced autonomic nervous system function (Cheng et al., 2023). An in vitro experiment showed that inhibiting the TLR2 signaling

pathway can suppress neuroinflammation and enhance the disease pathology of the central nervous system in PD (Daniele et al., 2015). Hence, the TLR2/MyD88/NF- κ B pathway might be a crucial mechanism in PD, and specific intervention in this pathway could offer novel approaches for preventing and treating PD.

Essential oil (EO) is extracted from natural plants, containing a high concentration of active ingredients and exhibiting remarkable pharmacological activity. The use of plant EOs for medicinal purposes dates back to ancient times. Notably, the Compendium of Materia Medica written by the Chinese medical scientist Li Shizhen, and Complete Herbal published by the British medical scientist Nicholas Culpeper both document the medicinal applications of various plant EOs. EO inhalation therapy is a notable supplementary treatment in traditional Chinese medicine, falling under the umbrella of conventional aromatherapy and nasal therapy. In this therapeutic approach, EO molecules are inhaled through the nasal passages, exerting preventive and therapeutic effects on various diseases. According to traditional Chinese medicine, the "nasal orifices connect to the brain". Recent medical research indicates that volatile components in essential oils can cross the blood-brain barrier and reach the brain by many pathways, including the olfactory nerve pathway, the olfactory mucosa epithelial pathway, and the bloodstream (Wei et al., 2019). This route of administration enhances drug targeting while minimizing drug toxicity and dosage, making it a potential mode of administration for brain diseases (Abd Rashed et al., 2021).

Several studies have shown the quick impact of essential oils in preventing, treating, and aiding in the recovery of nervous system disorders such as anxiety, depression, Alzheimer's disease, and PD (Kalilani et al., 2018). EOs are suggested as a possible approach to prevent and manage neurodegenerative conditions (Abd Rashed et al., 2021). Among these, the neuropharmacological activity of EOAT has been extensively investigated by scholars both domestically and internationally in recent years (Kim et al., 2015). *Acori Tatarinowii* Rhizoma is the first-class medicinal material in China's classic of materia medica, Shennong Materia Medica Classic. At the same time, it has the functions of clearing phlegm fire, calming liver wind, and relieving pulse arthralgia. The main active components of *Acori Tatarinowii* Rhizoma are volatile oils. α -Asarone and β -Asarone are significant elements of EOAT and have demonstrated a neuroprotective function in both in vivo and in vitro models of PD (Chellian et al., 2017). α -Asarone inhibits BV-2 microglial-induced neuroinflammation by inhibiting I κ B- α and I κ B- β and regulating NF- κ B transcription (Kim et al., 2015). β -Asarone, in combination with levodopa, modulates the activity of catechol-O-methyltransferase (COMT) and dopamine (DA) metabolism, leading to an enhancement in 6-OHDA-induced dyskinesia in PD rats (Huang et al., 2017).

Based on the aforementioned theory and pharmacological basis, this study aims to investigate the effectiveness and mechanism of EOAT inhalation in the treatment of PD. By utilizing a rotenone-induced PD rat model and an LPS-induced BV2 cell model, the role of EOAT in the PD model is examined. The findings of this study are expected to provide valuable insights and directions for the continued development and implementation of essential oil inhalation treatment in the context of PD.

2. Materials and methods

2.1. Main drugs and reagents

Rotenone (Med Chem Express, batch number: HY-B1756); Madopar (Shanghai Roche Led., batch number: YT0544); LPS(Sigma, batch number: L4391); DMEM high sugar medium(Procell, batch number: PM150210A); Fetal bovine serum(LONSA, batch number:S711-001S); Rat BDNF, IL-6, TNF- α , and IL-1 β , DA ELISA Kit (Calvin, batch number: E20230701A); Mouse IL-6 ELISA Kit (Calvin, batch number:CK-E20012M), TNF- α ELISA Kit(Calvin, batch number:CK-E20220M); The

Animal Tissue/Cell Total RNA Extraction Kit, the 2 × Universal Blue SYBR Green qPCR Master Mix (Servicebio, batch number: G3640), and the SweScript All-in-One RT SuperMix for qPCR (One-Step gDNA Remover) (Servicebio, batch number: G3337); TH Polyclonal antibody (Proteintech, batch number: 25859-1-AP); α -synuclein antibody (Servicebio, batch number: GB11773-100); Recombinant IBA-1 Polyclonal antibody (Servicebio, batch number: GB15105-100); Recombinant Anti-beta Actin antibody (Servicebio, batch number: GB15001-100).

2.2. Instruments and equipments

The following instruments were used in this study: the real-time fluorescence quantitative PCR instrument (Applied Biosystems, USA), the microplate reader (BIO-RAD Inc., USA), the gradient PCR instrument (Thermo Fisher Scientific, USA), the Rat Rod Rotator (Anhui Zhenghua Biological Instrument Equipment Co., Ltd., China), the OFT-100 rat and mouse autonomous activity device (Shenzhen Reward Life Science and Technology Co., Ltd., China), the ultra-micro spectrophotometer (Thermo Fisher Scientific, USA).

2.3. Plant material

Acori Tatarinowii Rhizoma was sourced from Wugong Mountain, Ji'an City, Jiangxi Province. The plant material was authenticated by Professor Chen Suiqing from the Department of Traditional Chinese Medicine Resources and Identification at Henan University of Traditional Chinese Medicine. The identification process aligned with the specifications outlined in the Pharmacopoeia of the People's Republic of China (Edition, 2020). A voucher specimen was deposited at the Pharmacology Laboratory of Henan University of Chinese Medicine, Zhengzhou, with the voucher specimen number for *Acori Tatarinowii* Rhizoma: 20230101 (January 01, 2023).

2.4. Animals

Male SD rats (SPF; 8 weeks old) (Animal qualification certificate No. 370726231100487935; Animal Ethics Approval No. DWLL202110056) were obtained from Jinan Pengyue Laboratory Animal Breeding Co., Ltd. (Certificate No. SCXK, 20220006; Shandong, China). The rats were housed in an SPF animal room (SYXK, 2021-00015) at the Experimental Animal Center of Henan University of Traditional Chinese Medicine. The environmental conditions in the room were maintained at a temperature of $(23 \pm 1)^\circ\text{C}$ and a humidity of $(55 \pm 15)\%$. The rats were provided with ad libitum access to food and water and were exposed to a 12-h light/dark cycle. All procedures performed in this study were following the guidelines of the Animal Care and Use Institutional Committee of Henan University of Traditional Chinese Medicine.

2.5. Preparation of EOAT

The EOAT was extracted following the standardized extraction method outlined in the Pharmacopoeia of China (2020 edition). To begin, the dried rhizomes of *Acorus calamus* Rhizoma were crushed and sieved using a No. 2 sieve. Precisely weighed amounts of calamus powder and distilled water (nine times the weight of the powder) were then mixed and soaked in an electric heating jacket for 1 h. Crushed ceramic stones were introduced into the mixture, and a volatile oil tester was connected. Distilled water was slowly added from the upper end of the tester until it reached the graduated portion, at which point it began to overflow into the flask. The addition of water was immediately stopped. Next, a reflux condenser tube was attached to the upper end of the tester, and the flask was heated on an electric jacket until boiling, thereafter maintaining a slight boil throughout the extraction process. Extraction was performed for a total of 5 h. Following the completion of extraction, heating was ceased, and the apparatus was allowed to cool for 30 min. Readings were then taken, and the EOAT was collected using

a sealable apparatus. The collected oils were subsequently dried overnight using anhydrous sodium sulfate, resulting in the production of a brown oily liquid. This liquid was stored in a brown bottle and placed in a refrigerator set at 4°C for backup purposes.

2.6. Identification of the constituents of EOAT

GC/MS system (Agilent Technologies, USA) and DB quartz capillary chromatography column ($30\text{ m} \times 250\text{ }\mu\text{m} \times 0.1\text{ }\mu\text{m}$) were used to analyze the constituents of EOAT. The testing conditions are as follows: carrier gas, helium (1 mL/min); split rate, 50:1. For the program heating conditions, the initial temperature was set at 60°C and gradually increased to 120°C at a rate of 8°C/min . It was then maintained for 2 min before being raised to 250°C at a rate of 2°C/min , also maintained for 2 min. The MS system utilized an EI ion source temperature of 200°C , a transmission line temperature of 250°C , an electron bombardment energy of 70 eV, and a bombardment voltage of 1.0 kV. The scanning range was set to 35–550 amu, and a solvent delay time of 3 min was implemented. The identification of the EOAT components was performed by comparing them with the mass spectrometry records in the National Institute of Standards and Technology (NIST) library in the United States. The proportion of each component was calculated using the area normalization approach.

2.7. Cell experiment method

2.7.1. CCK-8 assays

For BV2 cells, the CCK-8 method was used to screen for appropriate intervention dose of EOAT. BV2 cells in the logarithmic growth phase were seeded into 96-well plates and culture medium was added to the control wells. In the experimental groups, 100 μL of EOAT with mass concentrations of 1.0, 5.0, 10.0, 25.0, 50.0, and 100.0 $\mu\text{g/mL}$ were added to treat the cells respectively. After culturing for 24 h, CCK-8 reagent was added, and the absorbance (A) value at 450 nm was measured with a microplate reader to calculate the cell survival rate. Cell survival rate = (A experimental group - A blank group)/(A control group - A blank group). After preliminarily determining the dose of EOAT, a survival rate experiment was conducted to detect the cell survival rate of BV2 cells treated with EOAT after 24 h, 48 h, and 72 h.

2.7.2. Detection of inflammatory factors in cell supernatant

When BV2 cells in the logarithmic growth phase grow to a certain density, the experimental groups are intervened with 12.5, 25, and 50 $\mu\text{g/mL}$ EOAT respectively. The blank group and the LPS group are added with complete culture medium. After 2 h, except for the blank group, the other groups are all treated with 100 μL of LPS with a concentration of 1 $\mu\text{g/mL}$ (Qiu et al., 2022). After culturing for 24 h, the supernatant is collected and centrifuged, and the contents of TNF- α and IL-6 are detected according to the instructions.

2.7.3. IF detection of IBA-1 expression in BV2 cell

BV2 cells were treated with EOAT and LPS on a cover slide in a six-well plate and fixed, followed by treatment IF blocking solution at room temperature for 1 h. Subsequently, the cells were incubated with the primary antibodies overnight, followed by incubation with the corresponding secondary antibody for 1 h in the dark. Then, the cells were observed, and images were taken using a fluorescence microscope.

2.8. Animal method

2.8.1. Animals model preparation and drug intervention

After one week of adaptive feeding, the rats were randomly divided into the control group (Control), model group (Model), madopar group (Madopar, 50 mg/kg), high-dose EOAT group (EOAT-H, 100 $\mu\text{L/kg}$), medium-dose EOAT group (EOAT-M, 50 $\mu\text{L/kg}$), and low-dose EOAT group (EOAT-L, 25 $\mu\text{L/kg}$). Each group consisted of 12 rats. Except for

the control group, rats in each group were subcutaneously injected with rotenone solution ($1.0 \text{ mg kg}^{-1} \cdot \text{d}^{-1}$ for the first week, $1.5 \text{ mg kg}^{-1} \cdot \text{d}^{-1}$ for the second week, $2.0 \text{ mg kg}^{-1} \cdot \text{d}^{-1}$ for the third to fifth weeks, dissolved in 2 % DMSO sunflower oil) on the back of the neck once a day for 35 consecutive days to prepare the chronic PD model (Moradi Vastegani et al., 2023; Tieu, 2011). The Control was injected with the same amount of vehicle. Since the establishment of the model, 12 rats in the EOAT group were placed in a self-made inhalation apparatus ($50 \times 38 \times 15 \text{ cm}$) to sniff EOAT at 9 a.m. and 3 p.m. Beijing time every day. The EOAT and distilled water containing 1 % Tween-20 were ultrasonically emulsified at a ratio of 1:49, and then placed in an aroma lamp at the bottom of the olfactory box and heated for 30 min (Borgonetti et al., 2022). After the EOAT was fully diffused into the entire olfactory box, the rats were placed into the olfactory box to sniff twice a day for 1 h each time, continuously for 35 days. The Control and Model sniffed distilled water of equal volume, while the Madopar group took medication via gavage daily. Behavioral tests were performed one week before modeling, the first week, the third weeks, and the fifth weeks after administration (Fig. 1).

2.8.2. Behavioral test

2.8.2.1. General behavioral scoring (GBS). Following the administration, rats were evaluated using the behavioral scoring criteria established by Chen Xin et al. (Chen et al., 2008). The scoring scale ranged from 2 to 8, signifying successful modeling. Table 1 presents the detailed scoring criteria.

2.8.2.2. Rotarod test (RT). The rats were subjected to the fatigue rotating rod apparatus, where they were placed on the rotating rod. The rotation speed was set at 15 r/min, and the time from the beginning of rotation to the moment the rat fell off (falling latency) was measured. Each rat had three consecutive tests, separated by a 15-min gap. The average of the three results was considered as the final outcome. Before the actual testing, all rats underwent 3 days for adaptive training.

2.8.2.3. Open field test (OFT). Behavioral video recording analysis software was utilized to assess the independent locomotor activity of rats. An open box was prepared, measuring 50 cm in length, 50 cm in width, and 40 cm in height. The box's interior was coated in black, and the bottom was divided into 9 equally sized squares. Rats were introduced to the box from a consistent starting point, specifically the bottom right corner. Within a 5-min timeframe, the software recorded the rats' movement trajectory, distance traveled, and time spent at the starting point. Following each trial, the inner walls and floor of the open box were disinfected using 75 % ethanol before subsequent rats were placed inside for the continuation of the experiment.

2.8.3. Biochemical indicators detection

After 35 days of treatment, rats were fasted for 12 h and then

Table 1
General behavioral scoring criteria.

Scores	Scoring Criteria
1	Hair discoloration, dirt accumulation, back arching, raised hair, decreased resistance, decreased voluntary movement
2	1-Point presentation accompanied by a more pronounced decrease in voluntary movement, with tremor, slowed movement or unsteady gait.
4	Has a 2-point presentation with a markedly unsteady gait, or rotates to one side when walking, or has difficulty walking in a straight line
6	Reclining to one side, unilateral paralysis of anterior or posterior limbs, difficulty in eating and walking
8	Full unilateral paralysis of front or back limbs, with clonus in all four limbs, substantial weight loss, and inability to consume food
10	Completely unable to eat autonomously, dying or dead

weighed. Ten rats from each group were randomly chosen for dissection to collect the right striatum tissue homogenate 1 h following the last treatment. After centrifugation for 15 min, the supernatant was recovered from the homogenate. ELISA was used to measure the amounts of DA, BDNF, TNF- α , IL-1 β , and IL-6 in the striatum.

2.8.4. IHC staining to observe TH protein level in SN

Three rats were chosen at random from each group, and their SN areas were prepared for further study. The paraffin sections underwent standard procedures including roasting, dewaxing, and hydration. Dewaxing and dehydration were achieved by subjecting the sections to high pressure sterilization at 120 °C in citrate buffer with a pH of 6.0. The tissues were treated in 3 % hydrogen peroxide for 10 min to deactivate the natural catalase enzyme. The portions were then blocked with 10 % horse serum. Following this, overnight incubation with TH antibodies at 4 °C was carried out, followed by 1–2 h of incubation with HRP conjugated secondary antibodies at room temperature. Finally, the sections were incubated with streptavidin HRP. Regions with high concentrations of TH positive cells in the SN were chosen for microscopic examination at low magnification. The optical density of positive cell areas was calculated based on three high magnification fields of view, and the average was determined.

2.8.5. WB detection of TH protein level in striatum

Three rats were chosen at random from each group, and their left striatum tissues were weighed and combined with the tissue lysis solution. The tissue was homogenized at a low temperature and then lysed on ice for 30 min. The supernatant was recovered after centrifuging at 4 °C for 12,000 revolutions per minute for 15 min. The total protein content was assessed with the BCA technique. SDS-PAGE electrophoresis, membrane transfer, and blocking with 5 % skim milk powder for 1.5 h were then performed. The blocking solution was supplemented with TH antibody (diluted 1:2000), and the samples were left to incubate overnight at 4 °C. The samples were then washed three times with PBST at room temperature and exposed to a secondary antibody for 1 h. After three further rinses with PBST, ECL chemiluminescence was

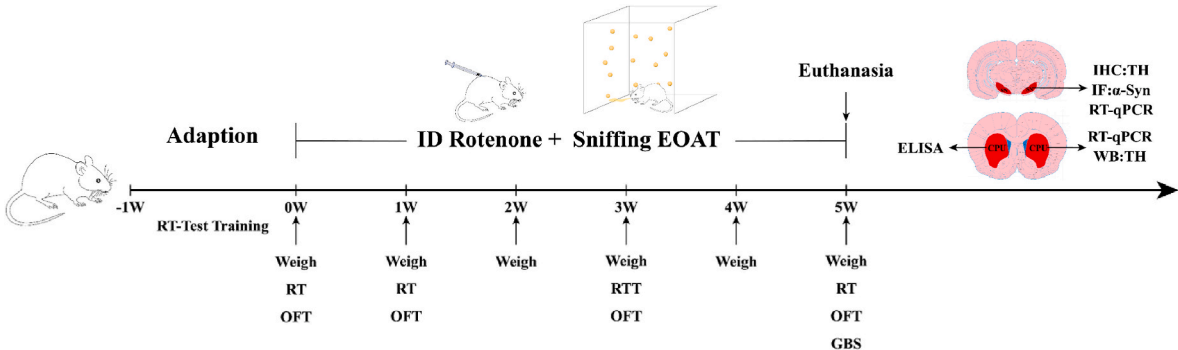


Fig. 1. Experimental processes.

produced.

2.8.6. IF detection of α -syn expression in SN

Three rats were chosen at random from each group and implanted in the midbrain area. Paraffin sections were dewaxed with water, followed by antigen retrieval and BSA blocking. Next, the slides were treated with a primary antibody, prepared in a specific proportion, and kept overnight at 4 °C. Subsequently, PBS washing was conducted for 5 min on each occasion. Subsequently, the tissues were exposed to secondary antibodies that matched the main antibodies at room temperature in the absence of light. Another round of PBS washing was carried out, followed by DAPI staining and further incubation in the dark. Finally, the slides were covered.

2.8.7. Detection of the mRNA levels in striatum and SN by RT-qPCR

Three rats were randomly chosen from each group, and the total RNA of the left striatum and SN were extracted according to the reagent kit's instructions. Later, 500 ng of total RNA was converted into cDNA using HyperScript RT SuperMix for qPCR. The relative expression was calculated using the $2^{-\Delta\Delta C_t}$ technique using HyperScript 2 × SYBR Green qPCR Master Mix for real-time PCR. The experiment aimed to quantify the mRNA levels of *IL-1 β* , *TNF- α* , *TH*, and *α -syn* in striatal tissue, along with *TLR2*, *MyD88*, *NF- κ B*, *IL-1 β* , *TNF- α* , *TH*, *α -Syn*, *Bcl-2* and *Bax* in SN. Refer to Table 2 for the primer sequence information.

2.9. Statistical analysis

Statistical analysis was performed using SPSS 21.0. Inter-group comparisons were conducted using a one-way analysis of variance (ANOVA). The data were provided as mean $\pm$ standard deviation ($\bar{x} \pm s$). The selection of either the LSD or Games-Howell test was based on the variance homogeneity. The immunohistochemical data were processed with ImageJ software and graphed using GraphPad Prism v.8.0 software. The threshold for statistical significance was established at a *p*-value below 0.05 ($P < 0.05$).

3. Results

3.1. The identification of major constituents of EOAT

GC-MS was employed to examine the composition of EOAT, and a total of 40 compounds were isolated. 25 compound names were identified, accounting for 96.06 % of the total compound content (Table 3, Fig. 3A). β -Asarone (69.54 %) and α -Asarone (19.97 %) in EOAT accounted for approximately 89.51 % of the total compound.

Table 2
The oligonucleotide primers (Rat).

Gene	Forward/Reverse	5'-3' primer sequence
<i>β-actin</i>	Forward	TGCTATGTTGCCCTAGACTTCG
	Reverse	GTTGGCATAGAGGTCTTTACGG
<i>TLR2</i>	Forward	CTCGGGACTCACAGCAAACAC
	Reverse	GGCATCTCCCTCTATTGTGTTGA
<i>MyD88</i>	Forward	CGAGGAGGACTGCCAGAAATAC
	Reverse	CAGTAGCAGATGAAGGCGTCG
<i>TH</i>	Forward	TCTGTGCGTCGGGTGTCTGA
	Reverse	GAATTGGTTACCCGTGCTTGTA
<i>TNF-α</i>	Forward	TACTGAACCTCGGGGTGATCG
	Reverse	AGAAGATGATCTGAGTGTGAGGGTC
<i>IL-1β</i>	Forward	TGACCTGTTCTTTGAGGCTGAC
	Reverse	CATCATCCCAGAGTCACAGAG
<i>Bax</i>	Forward	GGGTGGTTGCCCTTTTCTACTT
	Reverse	GAAGTCCAGTGTCCAGCCCAT
<i>Bcl-2</i>	Forward	CATGTGTGTGGAGAGCGTCAA
	Reverse	CAGCCAGGAGAAATCAAACAGA
<i>NF-κB</i>	Forward	GGATTCTCTGCTTACGGTGG
	Reverse	TTTCTTTCTCAGGGGGATTACAG

3.2. Effect of EOAT on the viability of BV2 cells

The CCK-8 results of EOAT on the survival rate of normal BV2 cells (Fig. 2A) show that when the concentration of EOAT reaches 100 μ g/mL, BV2 cells are significantly inhibited ($P < 0.01$), suggesting that this concentration is harmful to cells. Further survival experiments for 72 h found (Fig. 2B) that EOAT at concentrations of 12.5, 25.0, and 50.0 μ g/mL had no effect on the survival rate of BV2 cells. Therefore, concentrations of 12.5, 25.0, and 50.0 μ g/mL were selected for further research in subsequent experiments.

3.3. EOAT reduced the inflammatory level of BV2 cells

As shown in Fig. 2D and E, compared with the Control group, after LPS induction, the levels of TNF- α and IL-6 in the cell supernatant significantly increased ($P < 0.01$). Compared with the LPS group, different concentrations of EOAT can significantly reduce the levels of TNF- α and IL-6 ($P < 0.01$), and 50.0 μ g/mL EOAT can significantly down-regulate the level of TNF- α ($P < 0.05$). This indicates that EOAT can inhibit the release of inflammatory factors in BV2 cells.

3.4. EOAT decreased the expression of IBA-1 protein in BV2 cells

Fig. 2C shows that compared with the Control group, after LPS induction, the expression of IBA-1 protein in cells significantly increased ($P < 0.01$). Compared with the LPS group, different concentrations of EOAT can significantly reduce the expression of IBA-1 ($P < 0.01$). Among them, 50 μ g/mL EOAT can significantly down-regulate the level of TNF- α ($P < 0.05$), which is consistent with the change trend of inflammatory factors. This indicates that EOAT can inhibit the activation of BV2 cells and thereby reduce the occurrence of inflammation.

3.5. EOAT reduced the mortality and increased the body weight in the PD rat model

From the third week of modeling, deaths began to occur in the Model group of rats. After the administration ceased, the mortality rate in the Model group reached 16.67 % ($n = 10$). The mortality rates were 7.69 % in the Madopar, EOAT-H, and EOAT-L groups ($n = 11$). No animal fatalities were recorded in the Control and EOAT-M groups ($n = 12$). During the experiment, the body weight of rats in the Control group increased steadily and tended to be stable. Starting from the third week, the body weight of rats in the Model group dropped sharply and did not show a significant rebound until the end of the experiment. The changes in body weight of rats in the Madopar group, the EOAT-H, the EOAT-M, and the EOAT-L were similar to those of the Model group within 0–3 weeks. However, after 3–5 weeks, their body weight increased compared to that of the Model group. After the administration was completed, the body weight of rats in the Madopar group and the EOAT-H increased significantly compared to that of the Model group ($P < 0.01$), and the body weight of the EOAT-H increased significantly compared to that of the Model group ($P < 0.05$) (Fig. 3B).

3.6. Behavioral test

3.6.1. GBS

After one week of rotenone injection, the self-cleaning ability of rats in each group began to decline, with changes in fur color to yellow, slow gait, arched back, and reduced capture avoidance response. Behavioral scoring in the fifth week showed that all groups, except the Control group, had behavioral scores higher than 2, confirming the effective creation of a PD rat model in this study. The Model group exhibited a statistically significant rise in rat scores compared to the Control group ($P < 0.01$). The rat scores in each treatment group significantly dropped compared to the Model group ($P < 0.01$). After inhaling EOAT, the experimental results indicate that the rat model's apparent PD-like

Table 3
Compounds of the EOAT.

Number	Retention Time (min)	Compounds	Molecular formula	CAS	Area Sum/%
1	5.519	D-Camphor	C ₁₀ H ₁₆ O	464-49-3	0.04
2	5.904	L(–)-Borneol	C ₁₀ H ₁₈ O	464-45-9	0.56
3	6.033	(–)-Terpinen-4-ol	C ₁₀ H ₁₈ O	20126-76-5	0.04
4	6.302	cis-Anethol	C ₁₀ H ₁₂ O	104-46-1	0.56
5	6.451	2-(4-methylidenecyclohexyl)prop-2-en-1-ol	C ₁₀ H ₁₆ O	29548-13-8	0.21
6	9.346	(+)-Cyclosativene	C ₁₅ H ₂₄	22469-52-9	0.14
7	9.584	[1S-(1alpha,4alpha,7alpha)]-1,2,3,4,5,6,7,8-octahydro-1,4,9,9-tetramethyl-4,7-methanoazulene	C ₁₅ H ₂₄	514-51-2	0.06
8	10.405	(–)-Aristolene	C ₁₅ H ₂₄	6831-16-9	0.31
9	10.711	(+)-Calarene	C ₁₅ H ₂₄	17334-55-3	0.38
10	11.474	Methyl isoeugenol	C ₁₁ H ₁₄ O ₂	93-16-3	1.77
11	11.613	Octyl 4-methoxycinnamate	C ₁₈ H ₂₆ O ₃	5466-77-3	0.02
12	12.336	1H-Benzocycloheptene, 2,3,5,6,7,8,9,9a-octahydro-5,5,9-trimethyl-3-methylene-, (9S,9aS)-	C ₁₅ H ₂₄	60909-28-6	0.09
13	12.571	1-methyl-4-(1-methylethylidene)-2-(1-methylvinyl)-1-vinylcyclohexane	C ₁₅ H ₂₄	*3242-08-8	0.13
14	12.891	(–)-α-Gurjunene	C ₁₅ H ₂₄	489-40-7	0.10
15	13.107	(2S)-3α-Vinyl-3-methyl-2β-(1-methylvinyl)-6β-isopropylcyclohexanone	C ₁₅ H ₂₄ O	21698-44-2	0.15
16	13.577	Bicyclo[5.2.0]nonane, 4-ethenyl-4,8,8-trimethyl-2-methylene-	C ₁₅ H ₂₄	242794-76-9	0.19
17	14.957	β-Caryophyllene	C ₁₅ H ₂₄	87-44-5	0.03
18	15.108	Elemicin	C ₁₂ H ₁₆ O ₃	487-11-6	1.20
19	15.429	5H-Benzocyclohepten-5-one,6,7,8,9-tetrahydro-1,4-dimethoxy-	C ₁₃ H ₁₆ O ₃	79381-09-2	0.06
20	17.446	β-Asarone	C ₁₂ H ₁₆ O ₃	5273-86-9	69.54
21	17.977	Guaiene	C ₁₅ H ₂₄	88-84-6	0.11
22	18.971	Spiro[4.5]dec-6-en-8-one, 1,7-dimethyl-4-(1-methylethyl)-	C ₁₅ H ₂₄ O	39510-36-6	0.13
23	19.163	zierone	C ₁₅ H ₂₂ O	6754-66-1	0.25
24	19.682	α-Asarone	C ₁₂ H ₁₆ O ₃	2883-98-9	19.97

symptoms were relieved (Fig. 3C).

3.6.2. RT

One week post-modeling, the Model group of rats exhibited a significant reduction in the latency to fall off the rod compared to the Control group ($P < 0.01$). Rats in each treatment group exhibited a tendency towards longer delay to fall compared to the Model group ($P > 0.05$). During the third week of modeling, the Model group of rats showed a significant reduction in the time taken to fall off the rod compared to the Control group ($P < 0.01$). Rats in the EOAT-H and EOAT-M groups exhibited a notable rise in the latency to fall compared to the Model group ($P < 0.05$), whilst the remaining two groups shown a tendency towards higher latency ($P > 0.05$). During the fifth week of modeling, the rats in the Model group showed a noteworthy reduction in the time taken to fall off the rod compared to the Control group ($P < 0.01$). Rats in each treatment group exhibited a notable increase in the latency to fall compared to the Model group ($P < 0.05$). The findings indicate that EOAT can prolong the time it takes for rats to fall off the rod, enhancing the motor coordination of rats with PD (Fig. 4A).

3.6.3. OFT

Fig. 4D displays the rats' mobility paths in the OFT. The rats in the Model group exhibited decreased open field activity one week after modeling compared to the Control group ($P > 0.05$). Rats in each treatment group showed a trend of increased activity time compared to the Model group, with no statistically significant difference ($P > 0.05$). By the third week of modeling, the Model group showed a substantial reduction in open field activity time compared to the Control group ($P < 0.01$). Rats in the EOAT-H and Madopar groups exhibited a substantial increase in open field activity time compared to the Model group ($P < 0.05$), whereas the other two groups showed a tendency towards increased activity time ($P > 0.05$). After five weeks of modeling, the Model group of rats showed a decrease in spontaneous activity compared to the Control group, with a substantial reduction in movement duration in the open field ($P < 0.01$). The movement time of rats in each treatment group increased considerably compared to the Model group ($P < 0.01$) (Fig. 4B).

Fig. 4C displays the distances rats moved throughout during the OFT. Results indicated that the Model group of rats exhibited a notable reduction in movement distance compared to the Control group after one and three weeks of modeling ($P < 0.05$). The Madopar group showed a substantial increase in open field movement distance compared to the Model group ($P < 0.05$), but the EOAT group had a non-significant trend in movement distance ($P > 0.05$). The Model group of rats showed a notable reduction in movement distance in the open field compared to the Control group after five weeks of modeling ($P < 0.01$). The Madopar group had a notable rise in movement distance in the open field compared to the Model group ($P < 0.01$), while the EOAT-M group also shown a substantial increase in movement distance ($P < 0.05$). Both the high and low dosage EOAT groups exhibited an increasing trend in movement distance ($P > 0.05$). Olfactory exposure to EOAT can improve motor deficits in PD rats and increase their spontaneous activity.

3.7. EOAT regulates inflammatory factors, BDNF, and DA in rat striatum

Fig. 5A and B demonstrates a notable reduction in BDNF and DA levels in striatal of the Model group compared to the Control group ($P < 0.01$), suggesting impaired function of dopamine neurons releasing DA in the Model group. All treatment groups substantially elevated the levels of DA in striatal compared to the Model group ($P < 0.01$). The EOAT-M substantially boosted BDNF levels ($P < 0.05$), while the other treatment groups exhibited a non-significant increasing trend in BDNF levels ($P > 0.05$). Fig. 5C–E shows the levels of IL-1β, TNF-α, and IL-6 in striatal were considerably higher in the Model group compared to the Control group ($P < 0.01$). In the Madopar group, the levels of IL-1β, TNF-α, and IL-6 showed a lowering trend compared to the Model group, with no significant difference ($P > 0.05$). The EOAT-L group exhibited a notable reduction in IL-1β levels in striatal ($P < 0.05$), whereas TNF-α and IL-6 showed a declining trend ($P > 0.05$). The EOTA-M group showed a notable reduction in IL-1β levels in striatal ($P < 0.05$), a lowering tendency in TNF-α levels ($P > 0.05$), and a substantial decrease in IL-6 levels ($P < 0.01$). The EOAT-H group exhibited a notable reduction in IL-1β levels in striatal ($P < 0.05$), a substantial decrease in TNF-α levels ($P < 0.01$), and a considerable decrease in IL-6 levels ($P <$

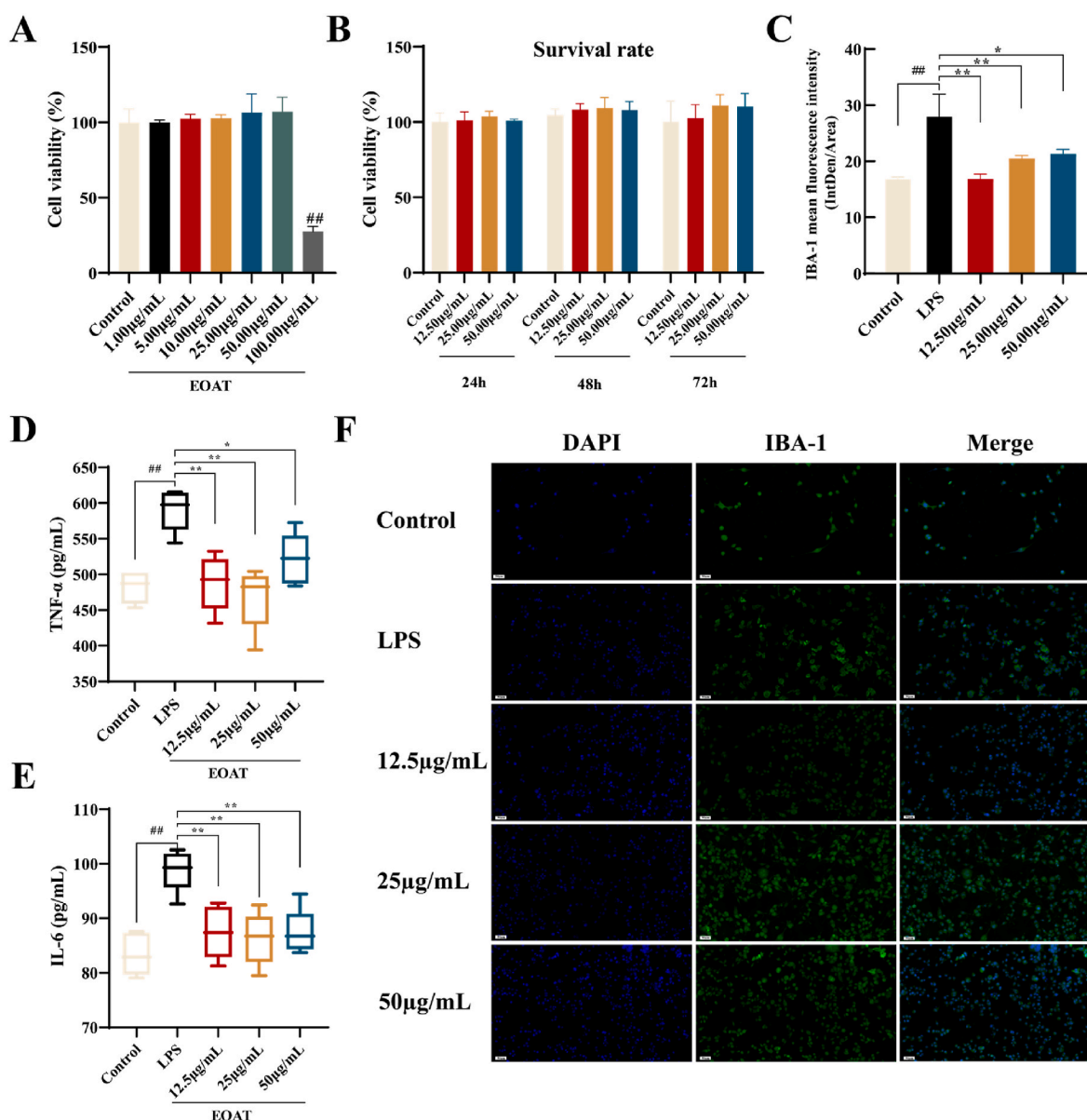


Fig. 2. EOAT relieves LPS-induced inflammation in BV2 cells. **(A)** Effects of EOAT with different concentrations on the viability of BV2 cells. **(B)** 72h survival rate experiment. $n = 6$ for the CCK-8 **(C)** Statistical results of the mean fluorescence intensity of the IF. $n = 3$ for the IF **(D)** IL-6; **(E)** TNF- α . $n = 6$ for the ELISA **(F)** The immunofluorescence results of IBA-1 protein in BV2 cells. The values are shown as the means $\pm$ SD. $\#P < 0.05$, $\#\#P < 0.01$ versus the control group; $*P < 0.05$, $**P < 0.01$ versus the model group.

0.05). The findings indicate that inhaling EOAT may have a neuro-protective impact via decreasing pro-inflammatory elements or elevating BDNF and DA levels.

3.8. EOAT increases the TH protein expression in SN by IHC

The TH-positive neurons in the SN are brownish-yellow in Fig. 5F and G. The Model group of rats exhibited a reduction in the quantity of TH-positive cells in the SN and a notable drop in optical density compared to the Control group ($P < 0.01$). The EOAT-H and EOAT-L groups exhibited a tendency of higher TH-positive expression in rats compared to the Model group ($P > 0.05$). Both the Medopar and EOAT-M groups exhibited a significant rise in TH-positive expression ($P < 0.01$).

3.9. EOAT increases the TH protein expression in SN by WB

IHC staining revealed that sniffing EOAT can enhance the expression of TH in SN. WB was conducted to assess the expression of TH protein in the striatum to elucidate the alterations of TH. Fig. 6A and B displays the results. The TH expression in the Model group considerably reduced compared to the Control group ($P < 0.01$). The Madopar group showed a substantial increase in TH expression compared to the Model group ($P < 0.05$). TH expression was considerably higher in the EOAT-H and EOAT-M groups ($P < 0.01$). The EOAT-L group had a tendency of elevated TH expression ($P > 0.05$).

3.10. EOAT decreases the α -syn over-activation in SN by IF

Fig. 6C and D demonstrated a substantial increase in the optical density of α -syn positive cells in SN of Model rats compared to the Control group ($P < 0.01$). The optical density of α -syn positive cells fell

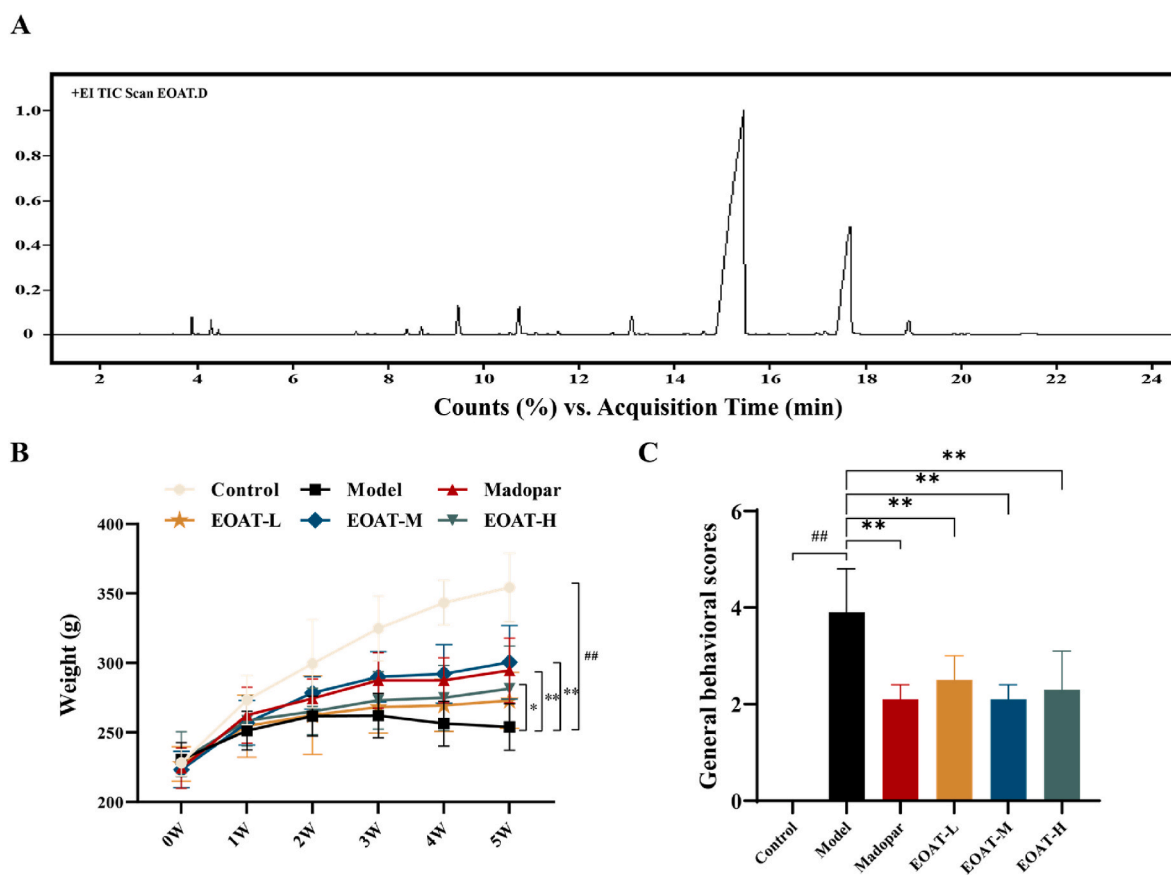


Fig. 3. The results of GC-MS and body weight of rats and general behavioral scores. **(A)** The results of GC-MS. **(B)** Body weight during the experiment. **(C)** General behavioral scores. The values are shown as the means $\pm$ SD. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ versus the control group; $^{*}P < 0.05$, $^{**}P < 0.01$ versus the model group, $n = 10$ – 12 for the behavioral test.

significantly in the Madopar, EOAT-H, EOAT-M, and EOAT-L groups compared to the Model group ($P < 0.05$). The findings of the IF indicate that inhaling EOAT can efficiently decrease the excessive activation of α -syn, leading to neuroprotective benefits.

3.11. EOAT regulates the mRNA levels of *IL-1 β* , *TNF- α* , *TH*, *α -syn* in striatum

RT-qPCR findings indicated a substantial rise in mRNA levels of *IL-1 β* , *TNF- α* , and *α -syn* in striatal of Model group rats compared to the Control group ($P < 0.01$). Conversely, the mRNA level of *TH* was dramatically lowered ($P < 0.01$). Following EOAT inhalation therapy, the Madopar group exhibited a decrease in mRNA levels of *IL-1 β* , *TNF- α* , and *α -syn* in striatal compared to the Model group ($P > 0.05$), along with an increase in *TH* ($P > 0.05$). The mRNA level of *TNF- α* considerably reduced in the EOAT-L group ($P < 0.05$). There was a lowering tendency in the mRNA levels of *IL-1 β* and *α -syn* ($P > 0.05$), and an increasing trend in *TH* ($P > 0.05$). The mRNA levels of *IL-1 β* and *TNF- α* in the EOAT-M group showed a substantial drop ($P < 0.01$), whereas *α -syn* declined dramatically ($P < 0.05$), and *TH* rose significantly ($P < 0.05$). In the EOAT-H group, the mRNA levels of *IL-1 β* and *TNF- α* showed a substantial drop ($P < 0.05$), while *α -syn* exhibited a declining trend ($P > 0.05$), and *TH* showed an increasing trend ($P > 0.05$). (Fig. 7A–D).

3.12. EOAT regulates the mRNA levels in SN

3.12.1. EOAT inhibits the expression of *TLR2/MyD88/NF- κ B* pathway mRNA in the SN of PD model rats

Fig. 7E–G displays the mRNA expression of the *TLR2/MyD88/NF- κ B* pathway in the rat SN. In the Model group, the mRNA levels of *TLR2*,

MyD88, and *NF- κ B* were considerably higher compared to the Control group ($P < 0.01$), suggesting strong activation of the *TLR2/MyD88/NF- κ B* pathway in the SN of model rats. In the Madopar group, the mRNA levels of *TLR2*, *MyD88*, and *NF- κ B* exhibited a declining trend compared to the Model group ($P > 0.05$). *TLR2* and *MyD88* were markedly reduced in all EOAT groups ($P < 0.01$). The mRNA expression of *NF- κ B* was considerably reduced in the EOAT-M group ($P < 0.05$) and exhibited a declining pattern in the EOAT-H and EOAT-L groups ($P > 0.05$). These findings indicate that EOAT can suppress the transcription of *TLR2/MyD88/NF- κ B* pathway mRNA.

3.12.2. EOAT inhibits the expression of *IL-1 β* , *TNF- α* , *α -syn* and *Bax* mRNA in the SN of PD model rats

Fig. 7H–K displays the mRNA expression of *IL-1 β* , *TNF- α* , *α -syn* and *Bax* in SN. The study's findings indicated a substantial rise in mRNA levels of *IL-1 β* , *TNF- α* , *α -syn* and *Bax* in the Model group compared to the Control group ($P < 0.01$), suggesting heightened inflammation and cell death in substantia of model rats. In comparison to the Model group, the mRNA levels of *IL-1 β* and *TNF- α* in the Madopar and EOAT-L groups exhibited a declining trend ($P > 0.05$), however *α -syn* and *Bax* were dramatically reduced ($P < 0.05$). *IL-1 β* , *TNF- α* , and *α -syn* were dramatically reduced in the EOAT-H and EOAT-M groups ($P < 0.01$), whereas *Bax* was significantly decreased ($P < 0.05$).

3.12.3. EOAT increases the expression of *Bcl-2* and *TH* mRNA in the SN of PD model rats

The expression of *Bcl-2* and *TH* mRNA in the substantia nigra of rats is shown in Fig. 7L and M. The results show that the mRNA levels of *Bcl-2* and *TH* are significantly decreased in the model group compared to the control group ($P < 0.01$). Compared to the model group, the mRNA

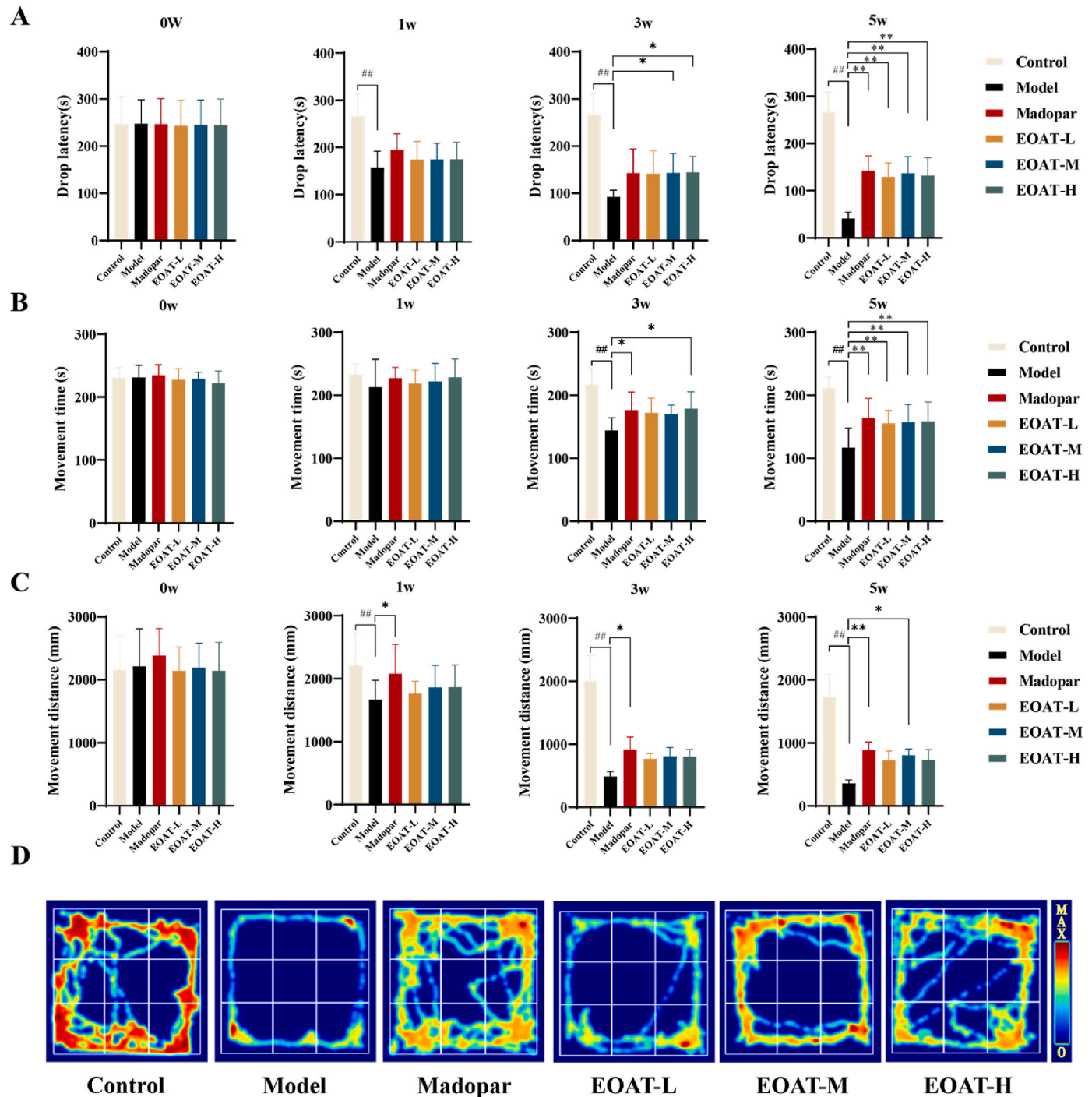


Fig. 4. Behavioral results. (A) Drop latency of RT. (B) Movement time of OFT. (C) Movement distance of OFT. (D) The mobility path diagram of OFT. The values are shown as the means $\pm$ SD. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ versus the control group; $^*P < 0.05$, $^{**}P < 0.01$ versus the model group, $n = 10-12$ for the behavioral test.

levels of *Bcl-2* and *TH* show an increasing trend in the Madopar and EOAT-L groups ($P > 0.05$). The mRNA level of *Bcl-2* is significantly increased in the EOAT-M group ($P < 0.05$), while the mRNA level of *TH* is significantly increased ($P < 0.01$). The mRNA level of *Bcl-2* shows an increasing trend in the EOAT-H group ($P > 0.05$), while the mRNA level of *TH* is significantly increased ($P < 0.01$). The RT-qPCR results suggest that EOAT can inhibit the activation of the TLR2/MyD88/NF- κ B pathway, alleviate neuroinflammation in the SNpc, and reduce the risk of stroke.

4. Discussion

PD, the second most prevalent neurological illness in the world, shows an exponential increase in prevalence with age (Ascherio and Schwarzschild, 2016). The global trend of aging is causing a significant rise in the number of PD patients globally, particularly in China. PD may be accompanied by depression, anxiety, cognitive impairment, olfactory impairment and other diseases in addition to motor dysfunction (Chen et al., 2021). These symptoms significantly impact both the quality of life and life safety. Furthermore, the condition has a protracted duration, and the intricate treatment and care regimen will impose significant

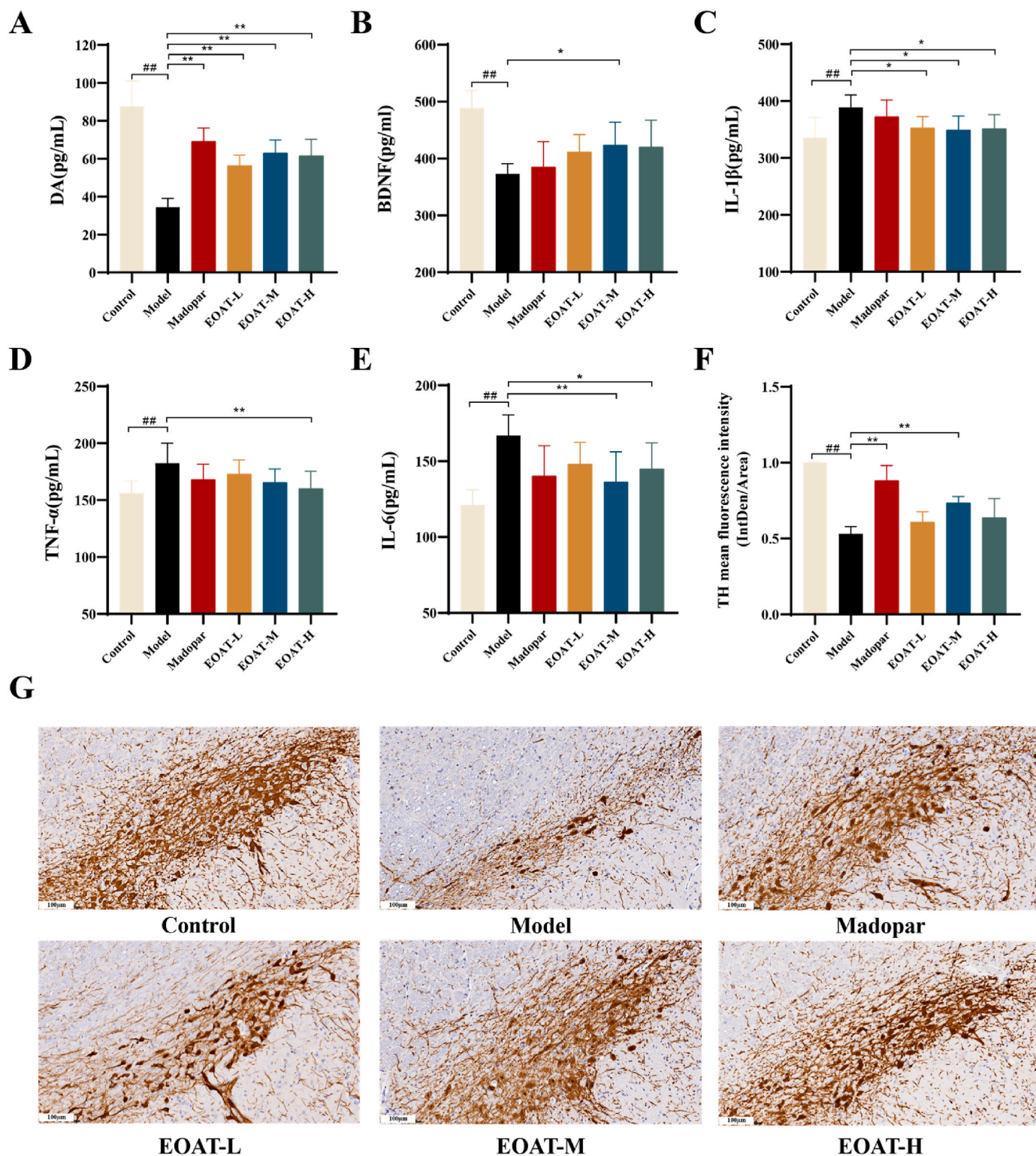


Fig. 5. The levels of neurotransmitters and inflammatory factors and IHC results of TH in SN. (A) DA; (B) BDNF; (C) IL-1 β ; (D) TNF- α ; (E) IL-6. $n = 7-9$ for the ELISA (F) Statistical results of the fluorescence intensity of the IHC, $n = 3$ for the IHC. (G) The IHC results of TH in SN. The values are shown as the means $\pm$ SD. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ versus the control group; $^*P < 0.05$, $^{**}P < 0.01$ versus the model group.

psychological and financial strain on the patient's family and community (Charvin et al., 2018). Therefore, it is of great social value and practical significance to explore new safe, effective and economical PD treatment strategies.

Acori Tatarinowii Rhizoma is the first-class medicinal material in China's classic of materia medica, Shennong Materia Medica Classic,

and it can be used for the treatment of coma and epilepsy. Modern research also indicates that the main active ingredient of *Acori Tatarinowii* Rhizoma is essential oil, which possesses excellent neuropharmacological activities. In this study, the components of the EOAT were analyzed by GC-MS. The components with the highest contents were β -Asarone, α -Asarone, and Elemicin, which is consistent with the

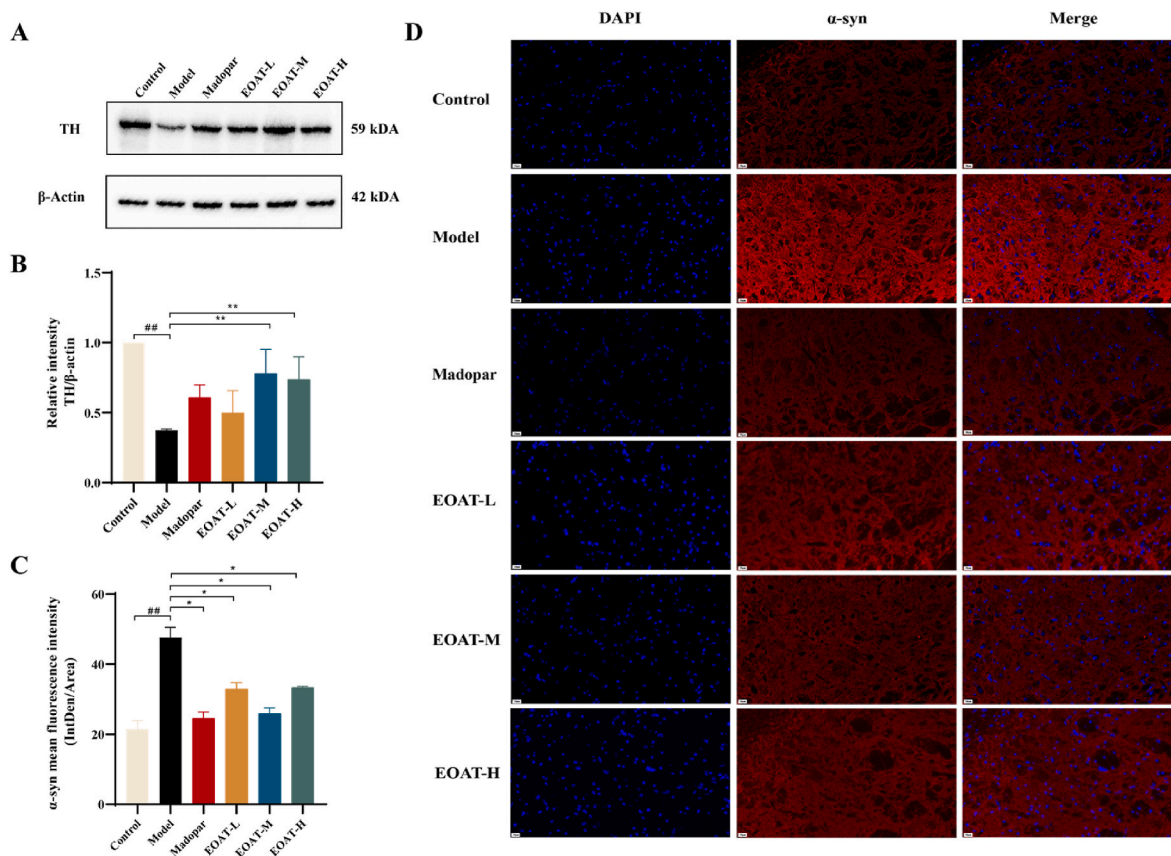


Fig. 6. The levels of TH and α -syn in SNpc. **(A)** The Western blot results of TH protein in SNpc. **(B)** Statistical results of the relative intensity of the WB. $n = 3$ **(C)** Statistical results of the mean fluorescence intensity of the IF. $n = 3$ **(D)** The immunofluorescence results of α -syn protein in SNpc. The values are shown as the means $\pm$ SD. $\#P < 0.05$, $\#\#P < 0.01$ versus the control group; $*P < 0.05$, $**P < 0.01$ versus the model group.

existing research findings (Liu et al., 2017). β -Asarone and α -Asarone have a wide range of neuropharmacological activities. β -Asarone can inhibit excessive excitation and neuroinflammation through the TLR4/NF- κ B/NLRP3 signaling pathway, thereby relieving neuropathic pain (Liu et al., 2024). α -Asarone can reduce the levels of inflammatory factors such as IL-6, IL-1 β , and TNF- α in rats with chronic sciatic nerve pain and repair damaged neurons (Zhang et al., 2023). A review study also shows that β -Asarone and α -Asarone can cross the blood-brain barrier, target and inhibit the activity of microglia, suppress the occurrence of neuroinflammation, improve the cognitive function of Alzheimer's disease mice, and exert neuroprotective effects (Stojanović et al., 2024). Elemicin can reduce the expression of 5-HT in the brains of stressed mice and improve the depressive behaviors of mice (Zhang et al., 2021). Based on this, the activities of these components in EOAT may serve as the material basis for its intervention in Parkinson's disease through anti-neuroinflammatory effects.

The activation of microglia is the main trigger of chronic inflammation in the occurrence process of PD. When microglia are overly activated, a large amount of inflammatory factors will be released. These inflammatory factors will cause damage to neurons and affect the synthesis and release of neurotransmitters. In this study, we induced the activation of BV2 cells by using LPS, and the expression of IBA-1 protein reflects this. Then, we used EOAT to intervene in LPS-induced BV2 cells and found that EOAT inhibited the activation of BV2 cells and reduced the expression of inflammatory factors and proteins.

Rotenone was subcutaneously injected into the back of the neck to successfully replicate the rat model of PD in this study. Rotenone is a lipophilic botanical pesticide found in the roots of certain plants in the leguminous family. It has the ability to cross the blood-brain barrier, enter cells, target mitochondria, enhance the release of ROS, and disrupt

dopamine metabolism by inhibiting the function of mitochondrial oxidative respiratory chain complex I (Borgonetti et al., 2022). Long-term subcutaneous injection of rotenone in small doses can induce persistent neuroinflammatory reaction in substantia nigra (Zhang et al., 2017), resulting in apoptosis and necrosis of cells, resulting in PD-like symptoms. Studies have found that more than 90 % of PD patients are closely related to environmental pollutants, especially rotenone exposure (Aarsland et al., 2021). This model not only simulates the main features of PD, but its progressive course is also similar to the onset of PD (Ibarra-Gutiérrez et al., 2023). Therefore, this model has high research value in the treatment of PD. Studies have shown that male SD rats are often selected as experimental animals in this type of model because of their high drug tolerance and low mortality in the model. The modeling cycle is generally 3–6 weeks. If the time is too short, it is difficult to simulate the chronic poisoning symptoms caused by environmental factors in PD patients (Wang et al., 2018; Yuan et al., 2016); if the time is too long, the mortality rate of rats will increase. The usual injection doses of rotenone are 1, 1.5, 2 and 2.5 mg/kg, of which subcutaneous injection of 2 mg/kg favours α -syn formation (Yuan et al., 2016). However, many studies have shown that high doses of rotenone subcutaneous injection may cause acute poisoning in animals. In order to reduce the acute toxicity of rotenone in the early stage of model establishment and ensure the replication of PD-like symptoms, male SD rats were selected to replicate chronic PD model by subcutaneous injection of rotenone at 1 mg/kg in the first week, 1.5 mg/kg in the second week, and 2 mg/kg in the third to fifth weeks. In this study, no animal died in the first two weeks, rats in the Model group died from the third week, and at the fifth week, one rat died in each of the Model group, Medoba group, EOAT-H group and EOAT-L group, and the GBS of the Model group rats was greater than 2, indicating that the PD animal model was

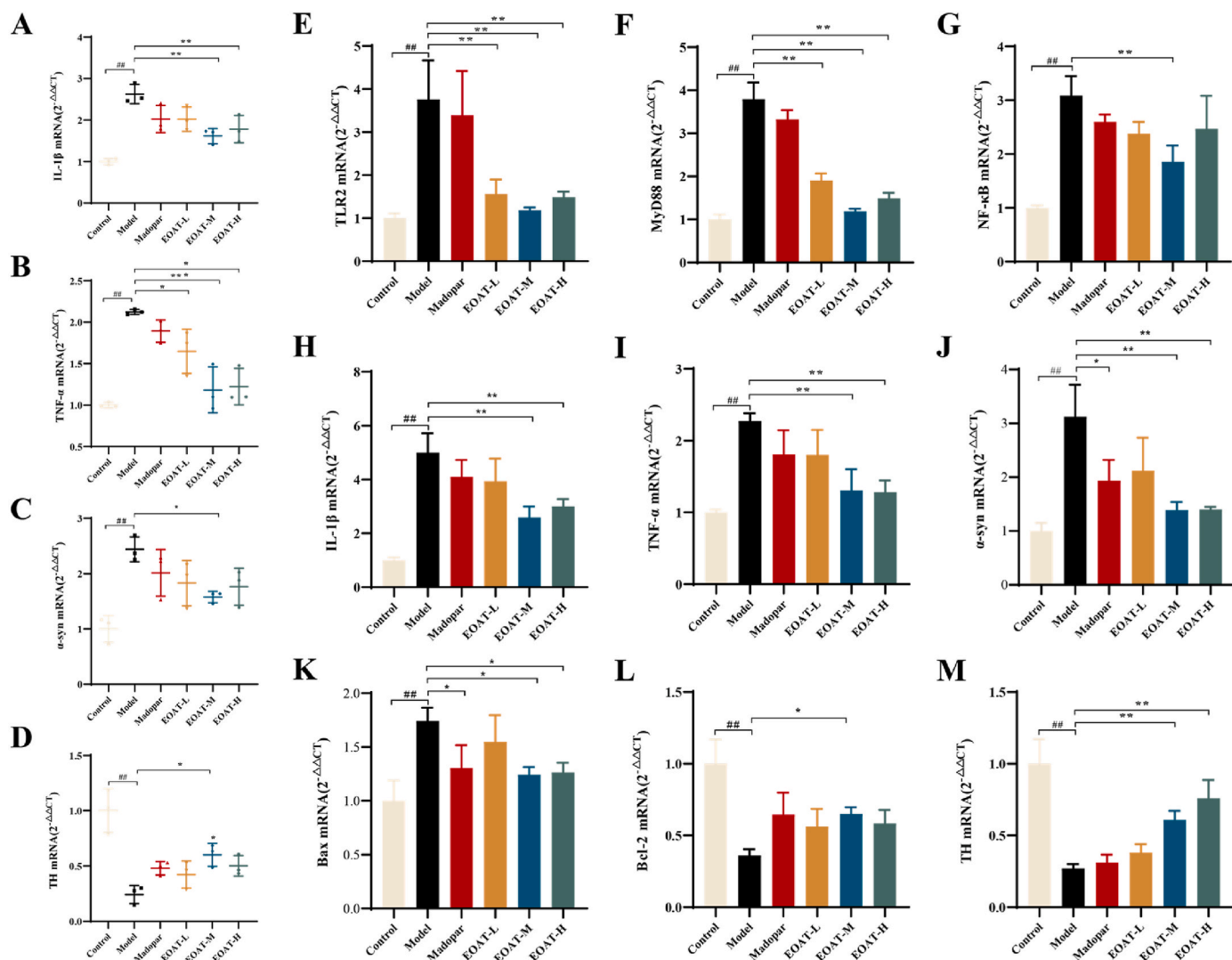


Fig. 7. The levels of the mRNA expression in striatum and SN tissues. (A–D) The mRNA expression of *IL-1β*, *TNF-α*, *α-syn*, *TH* in striatum tissues. (E–M) The mRNA expression of *TLR2*, *MyD88*, *NF-κB*, *IL-1β*, *TNF-α*, *α-syn*, *Bax*, *Bcl-2*, *TH* in SNpc. The values are shown as the means ± SD, *n* = 6. #*P* < 0.05, ##*P* < 0.01 versus the control group; **P* < 0.05, ***P* < 0.01 versus the model group.

successfully replicated. Animal deaths started happening in all treatment groups by the fifth week. Despite this, behavioral outcomes showed considerable improvement compared to the Model group, leading to the decision to conclude the animal experiment at the fifth week.

Animal behavior experiments, as an important means to comprehensively, comprehensively, and in real-time reflect the overall physiological and psychological status of animals, are considered the most reliable experimental method for studying the pathogenesis and protective measures of neurological and psychiatric diseases and functional disorders (Azman et al., 2022). PD, as a typical neurodegenerative disease, is characterized by four core clinical symptoms: bradykinesia, muscle rigidity, postural abnormalities, and resting tremors (McGregor and Nelson, 2019). In animal experiments, various behavioral experimental methods such as GBS, RT, and OFT are usually used to evaluate the improvement effect of drugs on animal motor function. The GBS is mainly based on observing the smoothness of animal hair, autonomous activity ability, mental state, reaction speed, movement posture, walking gait, tremor and paralysis, etc., to evaluate whether the animal model of PD has been successfully replicated. The RT aims to evaluate the motor function deficits of rodents, as they are prone to falling while running on the rotating rod when their motor coordination ability is

impaired. The length of the fall latency reflects the balance and coordination function of animals, and the shorter the latency, the poorer the animal's motor coordination ability (Shan et al., 2023). The OFT can be used to test the spontaneous activity of animals in a certain area within a specified time, and the distance and time of movement are the main reference indicators for evaluating autonomous activity (Lubrich et al., 2022). The low results of this experiment usually indicate a decrease in the animal's autonomous motor ability (Lubrich et al., 2022).

This study evaluated the PD like motor symptoms, motor function, and autonomous activity ability of rats using GBS, RT, and OFT. The experimental results showed that as the number of injections of rotenone increased, the hair smoothness of the Model group rats gradually decreased, the motor symptoms of slow gait and tremor paralysis gradually intensified, and the number of deaths increased. The GBS was significantly higher than that of the Control group. In addition, the fall latency period of the Model rats was significantly lower than that of the Control group, and the distance and time of exercise were significantly reduced, indicating that the modeling drug caused damage to the animal's motor function. After sniffing EOAT, the GBS of the Model rats decreased, the fall latency period was significantly prolonged, and the exercise time and distance increased, indicating that the motor function and autonomous activity ability of the model animals were improved.

In PD, degeneration and death of dopaminergic neurons in the substantia nigra striatum pathway can reduce DA synthesis and release, which in turn leads to motor dysfunction. DA, as a stress-responsive neurotransmitter and hormone (Speranza et al., 2021), is involved in regulating motor function. Under normal physiological environment, brain neurons synthesize and release DA through complex steps, and TH is a key rate-limiting enzyme that catalyzes DA synthesis reactions in organisms. BDNF is one of the most studied and characterized neurotrophic factors in the central nervous system, which affects synaptic plasticity and dopaminergic neurotransmission (Colucci-D'Amato et al., 2020; Nociti and Romozzi, 2023). The deficiency of TH and BDNF expression or function will directly affect the synthesis and secretion of DA, and then cause motor dysfunction (Azman and Zakaria, 2022). In this study, EOAT significantly upregulated TH and BDNF expression in nigrostriatal pathway of PD model rats, and exerted a regulatory effect on motor function by increasing DA synthesis and secretion in striatum.

Neuroinflammatory processes play a pivotal role in the pathology of PD (Zhang et al., 2022; Butovsky and Weiner, 2018). Previous studies have shown that activation of TLR2/MyD88/NF- κ B signaling pathway can trigger local neuroinflammation to induce prodromal symptoms of PD in rats (Cheng et al., 2023). In this study, we confirmed that TLR2/MyD88/NF- κ B pathway was activated in SNpc. Meanwhile the levels of proinflammatory factors IL-1 β , TNF- α , IL-6 in nigrostriatal pathway of PD model rats were significantly increased, indicating that neuroinflammation occurred in this model. After inhalation of EOAT, the TLR2/MyD88/NF- κ B pathway in the SNpc of PD model rats was inhibited, the inflammatory response in the nigrostriatal pathway was downregulated, and apoptosis was reduced (Fig. 8).

5. Conclusion

This study confirms that the main active component of EOAT is β -asarone, α -asarone and EOAT can inhibit LPS-induced over-activation of BV2 and reduce the occurrence of inflammation. In addition, inhalation of EOAT can improve motor dysfunction in a PD rat model, promote the synthesis and secretion of DA, reduce cell apoptosis, and downregulate the inflammatory response of the nigrostriatal pathway. And this process may be related to the TLR2/MyD88/NF- κ B pathway. Next, the regulatory role and anti-inflammatory activity of EOAT on the

TLR2/MyD88/NF- κ B pathway should be retrospectively examined in subsequent studies. And the delivery pathways of EO components in the brain should be clarified. It is expected that the relevant studies will provide a more solid theoretical foundation for the application of essential oil inhalation therapy and the development EOs of Chinese medicine.

CRediT authorship contribution statement

Yilong Hu: Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization. Yanan Zhao: Supervision, Methodology, Conceptualization. Zhiguo Mao: Methodology. Jingying Yang: Methodology. Baoling Huang: Methodology. Jinxin Miao: Writing – review & editing, Supervision, Conceptualization. Mingsan Miao: Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Ethics statement

The animal study was reviewed and approved by Henan University of Traditional Chinese Medicine and approved by Experimental Animal Center of Henan University of Traditional Chinese Medicine (SYXK, 2021-00015).

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Abbreviations

EOAT	Essential oil of <i>Acori Tatarinowii</i> Rhizoma
EO	Essential oil

(continued on next page)

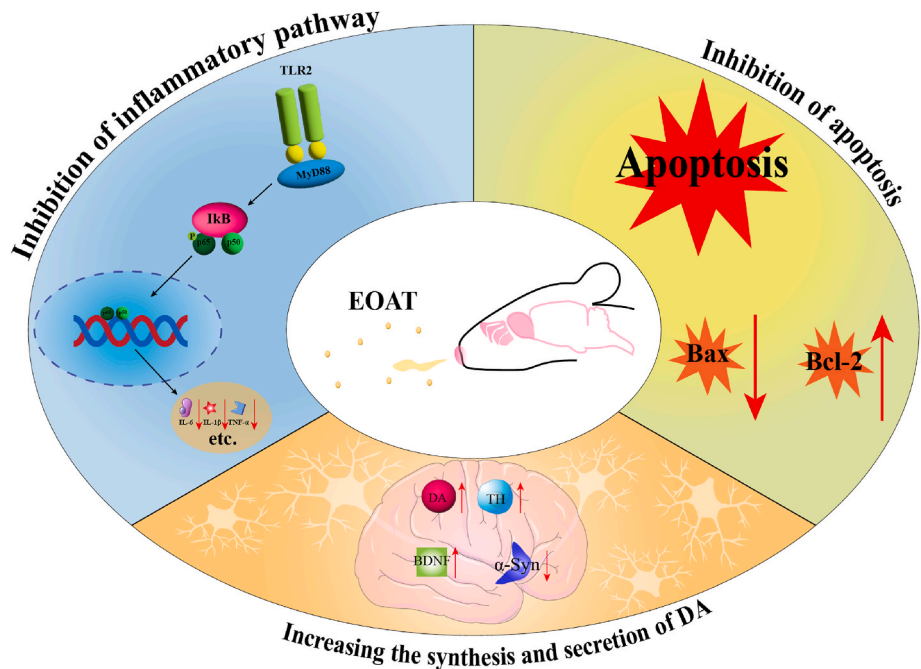


Fig. 8. Mechanism of EOAT sniffing on PD rat model.

(continued)

EOAT	Essential oil of <i>Acori Tatarinowii</i> Rhizoma
PD	Parkinson's disease
SD	Sprague-Dawley
ELISA	Enzyme-linked immunosorbent assay
IL-1 β	Interleukin1 β
IL-10	Interleukin10
TNF- α	Tumor Necrosis Factor- α
IL-6	Interleukin6
TGF- β	Transforming growth factor- β
BDNF	Brain-Derived Neurotrophic Factor
DA	Dopamine
IHC	Immunohistochemistry
SNpc	Substantia nigra pars compacta
SN	Substantia nigra
TH	Tyrosine hydroxylase
WB	Western Blot
α -Syn	α -Synaptic nuclear protein
RT-qPCR	Reverse transcription-polymerase chain reaction
GBS	General Behavioral Scoring
RT	Rotarod Test
OFT	Open Field Test
IF	Immunofluorescence
COMT	Catechol-O-methyltransferase
GC-MS	Gas Chromatography-Mass Spectrometry

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.119705>.

Data availability

Data will be made available on request.

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